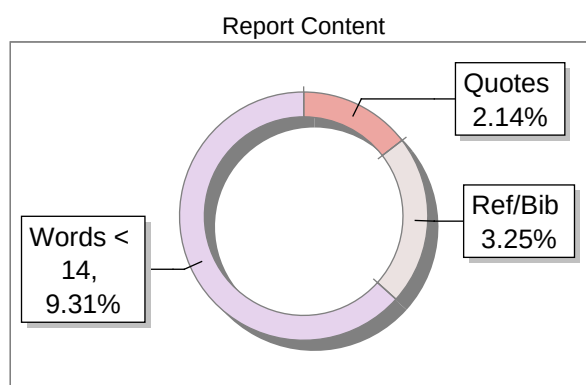
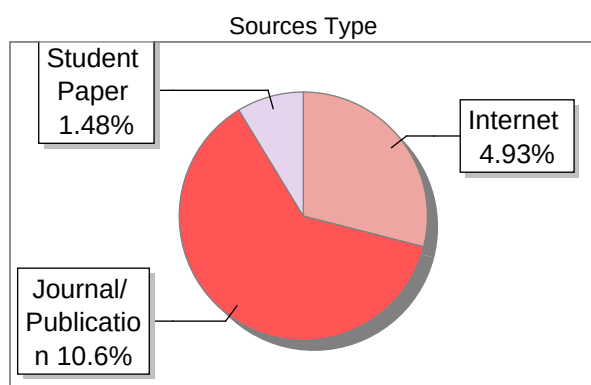


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Title	WEAPON DETECTIONANDALERT SYSTEM
Paper/Submission ID	5648605
Submitted by	lib@pesce.ac.in
Submission Date	2026-05-14 14:55:46
Total Pages, Total Words	48, 8933
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Project Report On

“WEAPON DETECTION AND ALERT SYSTEM”

Submitted in partial fulfilment for the award of Degree of

BACHELOR OF ENGINEERING

IN

INFORMATION SCIENCE AND ENGINEERING

Submitted by

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CERTIFICATE

This is to certify that, Kannika Rani B N [4PS22IS025], Thejas N [4PS22IS052], Uday R [4PS22IS053], Yashashwini P M [4PS22IS61] has been successfully completed the Project work "WEAPON DETECTION AND ALERT SYSTEM" in partial fulfillment for the award of Bachelor of Engineering of PES College of Engineering, Mandya (An Autonomous Institution Affiliated to VTU, Belagavi) during the year 2025-26. It is certified that all corrections indicated in internal assessment have been incorporated in the report deposited in the library. The Project Work have been approved as it satisfies the academic requirements in respect of work prescribed for the degree in Bachelor of Engineering in Information Science and Engineering discipline

Signature of the Guide Head of the Department

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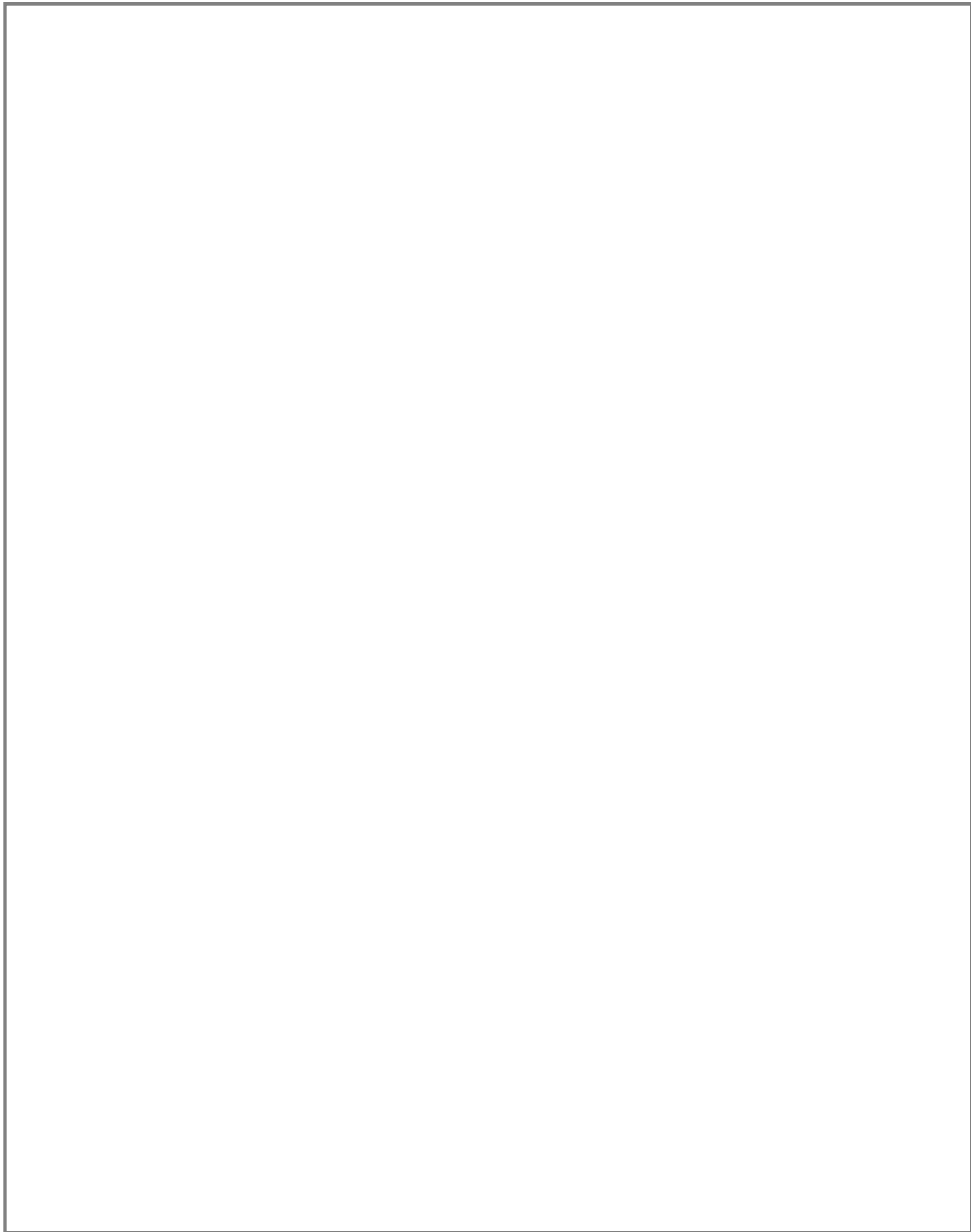
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DECLARATION

We, Kannika Rani B N [4PS22IS025], Thejas N [4PS22IS052], Uday R [4PS22IS053], Yashashwini P M [4PS22IS061], students of 8th Semester, Bachelor of Engineering in Information Science and Engineering at P.E.S. College of Engineering Mandya, hereby declare that the project work entitled “Weapon Detection and Alert System Using Deep Learning” is an authentic record of the work carried out by us during the academic year 2025–2026.

We further declare that this project has been undertaken by us independently under the guidance of our project guide, and it has not been submitted previously, either in part or in full, to any other university or institution for the award of any degree or diploma.

We also confirm that all sources of information used in this project have been properly acknowledged.

Date:

Place: Mandya

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ABSTRACT

The rise in weapon-related incidents across public spaces, educational institutions, and commercial environments has created an urgent need for intelligent surveillance systems capable of identifying threats in real time. Traditional security systems rely heavily on human monitoring, which is often limited by fatigue, delayed response times, and subjective judgment. To address these challenges, this project presents a Weapon Detection and Alert System using Deep Learning, designed to automatically identify weapons in live video streams and trigger immediate alerts to enhance situational awareness and response efficiency.

This system leverages the capabilities of modern deep learning architectures, specifically the YOLOv8 (You Only Look Once) family of real-time object detectors. In Phase 1 of development, focuses on knife detection, as knives are commonly used in close-range incidents and are challenging to identify using traditional surveillance systems. A curated dataset was collected, preprocessed, and used to train the YOLOv8 model. The system demonstrated strong performance with improving metrics such as precision, recall, and mean Average Precision (mAP), successfully detecting knives under varying lighting conditions, angles, occlusions, and complex backgrounds.

Phase 2 expands the system to detect multiple weapon types, including guns and other sharp objects. The dataset is extended and diversified to improve generalization, and the model is retrained to handle multi-class detection. This phase enhances the robustness and versatility of the system, allowing it to operate effectively in more complex real-world scenarios while reducing false negatives.

Phase 3 introduces an automated alert and notification system integrated with real-time detection. When a weapon is identified, the system triggers alerts such as on-screen warnings, email/SMS notifications, or integration with security dashboards. Additional features like timestamp logging, image capture of detected frames, and incident recording are implemented to support post-event analysis and evidence collection. This phase significantly improves response time and operational efficiency.

Phase 4 focuses on real-world deployment and optimization of the system to ensure practical usability. Techniques like model compression, quantization, and optimized inference are applied to maintain real-time performance with minimal resource usage. This phase also emphasizes scalability by enabling integration with large CCTV networks, cloud platforms, and centralized monitoring systems, making the solution suitable for smart city and large-scale security applications.

ACKNOWLEDGMENT

¹⁰³ We take this opportunity to express our sincere gratitude to ⁸⁹ all those who have supported and guided us throughout the successful completion of this project.

We are extremely thankful to our respected Principal, Dr. N. L. Murali Krishna, for providing us with a conducive academic environment and ⁹⁹ for his constant encouragement, which played a vital role in completing this project successfully.

¹ We express our heartfelt gratitude to Dr. T.S. Prabhakar, Associate Professor & Head of the ³⁸ Department of Information Science and Engineering, for his valuable guidance, constructive suggestions, and continuous support throughout the course of our project.

¹ We extend our sincere thanks to our Project Guide, Dr. T S Prabhakar, Associate Professor & Head, Department of Information Science and Engineering, for his constant supervision, technical guidance, and insightful suggestions. His encouragement, patience, and timely feedback helped us overcome challenges and complete the project efficiently.

We also thank the teaching and non-teaching staff of the Department of Information Science and Engineering for their cooperation and support during the project work. Finally, we express our gratitude to our friends and ⁹⁸ family members for their motivation and moral support throughout the project.

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CHAPTER 1

1.1 INTRODUCTION

Ensuring public safety in crowded and sensitive environments has become an increasingly critical challenge as incidents involving weapons continue to rise worldwide. Traditional security measures such as manual surveillance, fixed checkpoints, and human monitoring are often limited by fatigue, delayed response times, and the inability to continuously process large volumes of visual data. To overcome these limitations, intelligent automated surveillance systems have emerged as a powerful tool for proactive threat detection.

This report presents a Weapon Detection and Alert System that leverages deep learning to automatically identify weapons such as knives, guns, and other harmful objects in real time. By integrating advanced computer vision models with a responsive alert mechanism, the system aims to enhance situational awareness and significantly reduce reaction times during potential security threats. Using object-detection architectures such as YOLOV8 and optimized video-processing pipelines, the system can analyze live camera feeds with high accuracy and minimal latency.

The proposed solution not only reduces the burden on human operators but also strengthens preventive security measures in environments such as schools, airports, public transportation hubs, workplaces, and public events. Through intelligent automation, the system contributes to a safer environment by enabling early threat identification and rapid incident response.

The development of such a system aligns with the increasing demand for intelligent surveillance in areas where public safety is a priority. By combining deep learning techniques with automated alerting mechanisms, the proposed system enhances situational awareness, supports real-time decision-making, and contributes to a proactive approach to threat management. Moreover, the integration of advanced detection models ensures that the system remains robust across varying lighting conditions, angles, crowd densities, and background environments.

Overall, the Weapon Detection and Alert System represents a significant step toward modernizing security operations. It showcases how emerging technologies can be applied to create smarter, faster, and more dependable solutions for maintaining safety in critical environments. This report will further explore the problem addressed by the system, the objectives of the project, related research, the methodology used, and the overall performance of the implemented model.

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1.2 OBJECTIVES

The main objectives of the Weapon Detection and Alert System are as follows:

1.2.1 Knife detection and web cam integration

The primary objective of Phase 1 is to design, train, and validate an AI-based knife detection system using the YOLOv8 deep learning model. This phase forms the foundation for the larger project of real-time weapon detection and alert systems. The focus is not on building a complete safety system yet, but on creating a fully functional and accurate knife detection model that can detect the presence of a knife in images or video frames.

1.2.2 Weapon detection

Gun detection involves recognizing firearms such as pistols, revolvers, and rifles in video frames or images using models like YOLOV8. The system analyzes the shape, size, and features of guns to accurately detect them even in dynamic environments. Bats and clubs detection focuses on objects like baseball bats, wooden sticks, and metal rods that can be used as weapons. By integrating a webcam, the system captures live video streams, continuously processes each frame, and triggers alerts.

1.2.3 Real-Time monitoring and Alert system

The Real-Time Monitoring Weapon Detection and Alert System is designed to continuously observe an environment and identify weapons such as guns, knives, bats, or axes as they appear. Using deep learning models like YOLOV8, the system processes live video streams captured through a webcam or CCTV camera to detect and localize weapons in each frame. Real-time monitoring ensures that any weapon detected triggers instant alerts such as notifications, alarms, or messages to security personnel- allowing for immediate preventive action. The system analyzes each video frame continuously without delay, making it suitable for dynamic environments like schools, airports, offices, or public spaces.

Additionally, the system minimizes human intervention by automating surveillance tasks, thereby reducing the chances of human error and fatigue. The integration of real-time processing with an alert mechanism enhances situational awareness and ensures rapid response to potential threats. This makes the system highly effective for improving security and safety in real-world applications.

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Alert system

The Alert System in a Weapon Detection and Alert System is designed to notify security personnel immediately when a weapon is detected. Using deep learning models like YOLOV8, the system continuously analyzes live video streams from webcams or CCTV cameras to identify weapons such as guns, knives, bats, or axes. Upon detection, the system triggers instant alerts through notifications, alarms, or messages, enabling rapid preventive action. This ensures real-time monitoring and enhances safety in environments such as schools, airports, offices, and public spaces.

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1.3 PROBLEM STATEMENT

The current security infrastructure at public spaces such as ATMs, banks, and other high-risk facilities is inadequate to guarantee real-time threat detection and prevention. Despite the existence of CCTV surveillance, most systems rely on human operators to monitor video feeds — a process that is reactive, time-consuming, and subject to fatigue or human error. This leaves a critical gap: many weapons or concealed threats go undetected until after a crime occurs.

Further, the automated weapon-detection systems developed so far face significant limitations. They often lack real-time processing capabilities, struggle with occluded or partially visible weapons (e.g., guns or knives hidden by clothes, hands, or bags), and are prone to high false-positive rates — mistakenly flagging benign objects like phones or keys as weapons. In addition, most such systems are trained on small, narrow, or non-diverse datasets, failing to account for real-world variability such as different lighting conditions, camera angles, and diverse weapon types. Even where detection algorithms show promising accuracy in controlled environments, they rarely undergo rigorous validation in live, real-world surveillance contexts.

As a result, current security systems remain unreliable, unable to provide timely alerts or capture evidence effectively, which undermines their effectiveness in preventing violence and protecting public safety.

CURRENT CHALLENGES

Developing a model specifically for detecting knives in real-time from a webcam presents several challenges. One major issue is the limited availability of high-quality, labeled datasets containing diverse knife images in different lighting, angles, and occlusions. Another challenge is accurately detecting knives in cluttered or dynamic environments, where the object may be partially hidden, overlapping with other objects, or viewed from unusual angles. Real-time detection also demands efficient computation, as processing video streams continuously can strain hardware, particularly on edge devices or standard webcams. Additionally, high false-positive rates can occur due to objects resembling knives, such as kitchen utensils or tools, which can reduce the model's reliability. Finally, varying lighting conditions, camera resolutions, and motion blur further complicate detection, requiring the model to generalize effectively to different real-world scenarios.

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CHAPTER-2

2.1 LITERATURE REVIEW

Weapon detection has become an increasingly important research area due to rising security concerns in public spaces, transportation hubs, and surveillance systems. Early work in this domain relied heavily on traditional image-processing techniques, but recent studies have shifted toward deep learning-based solutions that offer higher accuracy and real-time performance.

[1]. M. T. Bhatti et.al, "Weapon detection in real-time CCTV videos using deep learning," in Proc. IEEE Virtual Reality Conference, 2021.

This work presents a deep-learning-based weapon detection system for real-time CCTV footage. The authors employ CNN models to detect weapons under dynamic surveillance conditions. Emphasis is placed on real-time processing speed and accuracy in crowded scenarios. Results show improved detection reliability compared to traditional vision systems.

[2]. Chadha et.al, "Low cost concealed weapon detection for schools using acoustic signatures," in Proc., IEEE Int. Conf. Pattern Recognition (ICPR), 2019.

The study explores low-cost concealed weapon detection using acoustic signatures. It uses sound- pattern analysis rather than visual inputs to identify potential threats. The approach is designed to be affordable for school environments. Experiments demonstrate promising accuracy in classifying weapon- related acoustic cues.

[3]. A. Goenka et. al, "Weapon detection from surveillance images using deep learning," in Proc. IEEE ICNWC (2nd Int. Conf. Networking & Communications), 2022.

This paper applies deep learning to weapon detection in surveillance images. The authors use advanced CNN architectures to classify and localize weapons. The system is tested on The system is tested on diverse image datasets to enhance robustness. Findings highlight improved performance compared to earlier CNN models.

[4]. S. N. Bushra et al, "Smart video surveillance based weapon identification using YOLOv5, in Proc. IEEE Int. Conf. Electronic Systems (ICESI), 2022

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The authors propose a YOLOv5-based smart surveillance system for weapon identification. YOLOv5 enables high-speed detection suitable for real-time security applications. The study addresses challenges like occlusion and low-resolution footage. Results show significant gains in precision and recall across varied scenarios.

[5]. V. Sa-ing et.al, Weapon detection method using convolutional neural network with human pose estimation, in Proc. IEEE, 2025.

This work integrates CNNs with human pose estimation for weapon detection.

The method enhances context-aware detection by analyzing human posture. It improves recognition accuracy when weapons are partially concealed. Evaluation demonstrates strong performance in complex surveillance environments.

[6]. R. L. Sikka et.al, "Real-time weapon detection through deep learning techniques," IEEE, 2024.

The paper introduces a real-time deep learning approach to weapon detection. It focuses on optimizing neural network architectures for fast inference. The system is evaluated on real-world surveillance datasets. Results confirm reliable detection under varying lighting and background conditions.

[7]. J. G. B. Patturose et al, "Pixels to protection: An intelligent weapon detection system," IEEE, 2024.

This study proposes an intelligent weapon detection system titled "Pixels to Protection." It uses deep neural networks to extract discriminative visual features. The emphasis is on automated threat response and continuous monitoring. Experiments indicate high accuracy and strong potential for security deployment.

[8]. H. Tabassum et al, "Performance analysis on enhancing security through weapon detection," IEEE, 2024.

The paper analyzes performance improvements in security systems via weapon detection. Several deep-learning models are compared for efficiency and accuracy. The study examines factors affecting real-time detection reliability. Findings provide insights for optimizing surveillance-based security frameworks.

[9]. S. Singh, A. Verma and R. Kumar, "The vanguard of weapon detection via CNN-SVM, IEEE, 2024 .

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The authors introduce a hybrid CNN–SVM architecture for weapon detection. CNNs extract deep visual features, while SVMs perform final classification. This hybrid method improves precision over standalone CNN models. Experiments demonstrate strong consistency across varied image backgrounds.

[10]. D. X. Nguyen et al, "Automatic weapon detection using deep learning," IEEE, 2023.

This research explores automatic weapon detection using deep learning. Advanced CNN architectures are utilized for object recognition tasks. The system targets real-time applicability in surveillance networks. Results show enhanced accuracy and robustness against environmental noise.

[11] M. Abdullah et al, H. Khan, and S. Ahmed, "Advancing small-sized handgun detection: Challenges and solutions," IEEE, 2024.

The study addresses challenges in detecting small-sized handguns. It proposes specialized deep-learning techniques to handle tiny-object detection. Key issues like low visibility and object scale are thoroughly examined. The results demonstrate notable improvements in detecting miniature firearms.

[12] V. S. Rai, R. Sharma, and A. Patel, "Gun detection system using YOLO v3," IEEE, 2019.

This work develops a gun detection system using YOLOv3. YOLOv3 enables fast detection suitable for real-time surveillance applications. The study evaluates detection under varying illumination and angles. The system achieves reliable accuracy for practical security scenarios.

[13]. A. Jain et.al "IEEE weapon detection paper (YOLOv8-based real-time system), in Proc. IEEE Conference Publication, 2025.

The paper presents a YOLOv8-based real-time weapon detection system. YOLOv8 enhances speed, accuracy, and small-object detection performance. The methodology emphasizes deployment readiness for live CCTV feeds. Tests show superior results compared to earlier YOLO versions.

[14]. A. S. Mandal et.al "Weapon detection using artificial intelligence and deep learning," IEEE, 2020.

This study provides an AI and deep-learning-based approach to weapon detection. It uses convolutional networks to detect and classify firearms in images. Focus is placed on reducing false alarms in surveillance systems. Where

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Results demonstrate improved efficiency relative to conventional methods.

[15]. F. A. Khan et. al, ²⁸ A deep learning-based efficient firearms monitoring technique for building secure smart cities" IEEE Access, 2023.

The authors introduce a deep-learning-based firearm monitoring technique for smart cities. The approach is optimized for large-scale, real-time urban surveillance. It enhances detection accuracy through efficient neural network design. The study confirms scalability and practical applicability across city infrastructures.

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2.2 GAP IDENTIFICATION

2.2.1 Lack of Real-Time Performance: Many ⁹⁶existing detection systems are either not fully automated or are computationally intensive, preventing instant threat identification and alerting in live video feeds. Some models require offline processing or high-end GPUs, ⁸⁰which makes them unsuitable for deployment in real-time surveillance applications, especially in crowded public spaces where immediate action is critical.

2.2.2 High Rate of False Positives: Current systems frequently misclassify harmless objects such as mobile phones, keys, wallets, or metallic items as weapons. These false alarms reduce the overall reliability of the system and can overwhelm security personnel, causing delayed or inappropriate responses. Reducing false positives remains a significant challenge for practical deployment.

2.2.3. Dataset Quality and Diversity: Many studies rely on small, custom, or non-diverse datasets, which fail to capture the wide variety of real-world scenarios. Most datasets do not comprehensively cover different weapon types, including firearms, knives, and improvised weapons, nor do they account for variations in lighting, camera angles, resolution, occlusion, or crowded environments. This lack of diversity limits the ability of models to generalize effectively in real-world surveillance conditions.

2.2.4. Dependence on Environment and Hardware: Several existing models require high-end ²⁵hardware or optimized environments for real-time processing, which makes them unsuitable for low-power edge devices such as CCTV cameras or mobile surveillance units. This dependency limits the scalability and practicality of automated detection systems in cost-sensitive or resource-constrained settings.

2.2.5. Limited Real-World Validation: Many weapon detection models are tested only on benchmark datasets or simulated environments. They often lack rigorous validation in real-world scenarios over extended periods, leaving questions about their long-term reliability, robustness, and performance in dynamic, crowded, or unpredictable environments.

2.2.6. Poor Model Interpretability: ⁵⁷Deep learning models often function as black boxes, ²²making their decision processes hard to understand. This lack of interpretability can lead to failures in unexpected situations, such as new weapon designs, unusual object handling, or changes in public behavior, reducing operator trust.

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2.2.7. Occlusion and Concealment: Detecting weapons that are partially hidden remains a critical challenge. Objects concealed under clothing, bags, or hands are difficult for even advanced models to recognize. Occlusion reduces detection accuracy, increasing the risk of missed threats and limiting the effectiveness of automated surveillance in crowded or complex environments.

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CHAPTER-3 (SYSTEM REQUIREMENTS AND ANALYSIS)

3.1 Software Requirements

The software requirements necessary for ¹⁰⁴the development and deployment of the Weapon Detection and Alert System are as follows:

Operating System:

Windows or Linux provides the platform to run and manage the system efficiently. ³Ubuntu is preferred for better compatibility with deep learning frameworks.

Programming Language:

⁷⁹Python is used due to its simplicity and strong support for AI/ML libraries. It enables quick development and easy integration of models.

Development Environment:

Tools like VS Code, PyCharm, or Jupyter Notebook are used to write, test, and debug code. They improve productivity with features like auto-completion and debugging.

Libraries & Frameworks:

OpenCV handles image/video processing, while NumPy and Pandas manage data. PyTorch/TensorFlow and YOLOv8 are used for building and running the detection model.

Annotation Tools:

LabelImg and Roboflow help in labeling images for training the model. They ⁷⁹create bounding boxes around objects like weapons.

Database (Optional):

MySQL or Firebase can store alert logs and detection records. This helps in tracking and analyzing system activity.

APIs & Services:

SMTP is used to send email alerts, and Twilio API sends SMS notifications. These ensure immediate communication when a weapon is detected.

Version Control:

Git is used to manage code changes and track project progress. It helps in collaboration and prevents loss of work.

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3.2 Hardware Requirements

Processor:

A powerful processor like Intel i5 or Ryzen 5 ensures smooth execution of the detection model. It helps in faster data processing and real-time performance.

RAM:

Minimum 8 GB RAM is required to handle model loading and video processing. Higher RAM (16 GB) improves speed and prevents system lag.

Storage:

At least 256 GB storage is needed to store datasets, models, and project files. SSD is preferred for faster data access and performance.

GPU:

An NVIDIA GPU accelerates deep learning tasks like training and inference. It significantly improves processing speed compared to CPU.

Camera:

A webcam or CCTV camera is used to capture real-time video input. It acts as the primary source for detecting weapons.

Internet Connectivity:

Internet is required to send alerts via email and SMS services. It also helps in accessing APIs and cloud-based resources.

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CHAPTER-4 (OBJECTIVE 1)

4.1 INTRODUCTION

Knife detection and web cam integration

In today's world, ensuring safety in public and private spaces has become increasingly critical. One of the major threats to security is the presence of weapons, particularly knives, which can be used in violent incidents. Detecting knives in real-time can play a crucial role in preventing harm and ensuring rapid response from security personnel. Knife detection is a specialized area of computer vision that focuses on identifying knives in images or video frames. Using advanced deep learning techniques, such as object detection algorithms like YOLOV8 (You Only Look Once), it is possible to train systems to recognize knives accurately based on their shape, size, and visual features.

Integrating knife detection with a webcam allows the system to operate in real-time, continuously monitoring an environment for potential threats. The webcam acts as the input device, capturing live video streams that are processed frame by frame by the detection model. This real-time analysis ensures that any knife appearing in the camera's field of view is immediately detected, and alerts can be triggered without delay. The combination of knife detection and live video processing not only improves the accuracy of detection but also enhances the practical usability of the system in dynamic and crowded environments such as schools, airports, offices, and public spaces. This capability allows security teams to respond promptly, preventing potential accidents or crimes before they escalate. The system also reduces the dependency on human monitoring, which can be error-prone and less efficient in high-traffic areas.

Moreover, the integration of webcam-based detection with deep learning models allows the system to adapt to varying lighting conditions, angles, and backgrounds, making it robust in real-world scenarios. Continuous updates and training of the detection model can further improve accuracy and reduce false positives, ensuring reliable performance over time. The system can also be combined with additional sensors or security frameworks to create a comprehensive safety ecosystem, enabling smarter decision-making and predictive threat management.

By achieving this objective, the system provides a foundation for automated security monitoring, reducing the dependence on human surveillance and enabling rapid preventive measures. Knife detection and webcam integration serve as the first critical step in a broader Weapon Detection and

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Alert System, forming the core of real-time threat identification and proactive safety management. This objective emphasizes the importance of using modern technology to enhance public safety, minimize response time, and create a robust, intelligent security infrastructure.

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4.2 PROPOSED METHODOLOGY

The proposed methodology outlines the step-by-step approach for developing a real-time knife detection system integrated with a webcam, using deep learning techniques. The methodology is divided into several key phases:

4.2.1 Data collection

Gather a comprehensive dataset of knife images from multiple sources, including publicly available datasets and manually collected images.

Ensure diversity in images, covering different types of knives, angles, lighting conditions, and backgrounds.

Label the dataset with bounding boxes to indicate the knife locations, which will be used for supervised learning.

4.2.3 Data Preprocessing

Resize images to a uniform dimension to maintain consistency during model training.

Apply data augmentation techniques such as rotation, flipping, scaling, and brightness adjustment to make model more robust

Normalize image pixel values to improve model convergence during training.

4.2.3 Model Selection and Training

Choose an object detection model suitable for real-time detection, such as YOLOv8.

Train the model on the preprocessed dataset to detect knives, adjusting hyper parameters to optimize accuracy and speed.

Evaluate the model using metrics like precision, recall, and mean Average Precision (mAP) to ensure reliable detection.

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4.2.4 Webcam Integration

Use a webcam or CCTV camera as the real-time video input device.

Capture video frames continuously using a programming library like OpenCV in Python.

Preprocess each frame as per the model's input requirements and feed it to the trained knife detection model for analysis.

4.2.5 Real-Time Knife Detection

The model analyzes each frame to identify knives and **draws bounding boxes around detected** objects.

Implement confidence thresholds to filter out false detections and improve accuracy.

4.2.6 Testing and Optimization

Test the integrated system in real-time scenarios under different lighting, angles, and environments

Optimize model performance to ensure minimal delay and high detection accuracy.

Fine-tune the system for real-world deployment in schools, airports, offices, or public spaces.

4.2.7 Figure (Flow Chart)

The flowchart shows the steps of a real-time knife detection system. First, knife images are collected, labeled, resized, and preprocessed. Then a deep learning model (like YOLOV8) is trained and integrated with a webcam to analyze live video frames. Finally, the system detects knives in real time, draws bounding boxes, and is tested and optimized for accuracy and speed.

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4.1 Figure of Proposed Methodlogy Flowchart for Real-Time Detection System

Description of Flow Chart

The flowchart illustrates the workflow of a real-time knife detection system. It begins with collecting and preprocessing knife images, including labeling and resizing the data. A deep learning model such as YOLOV8 is then trained and connected to a webcam to process live video frames. Finally, the system identifies knives in real time, displays bounding boxes, and is evaluated and optimized for accuracy and performance.

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4.3 RESULT AND IMAGES

RESULT

The Knife Detection and Webcam Integration system was successfully implemented and tested in real-time scenarios. The system was able to capture live video feed from the webcam and process each frame to detect knives accurately. Using a deep learning object detection model like YOLOV8, the system identified knives in various lighting conditions, angles, and backgrounds. Bounding boxes were drawn around detected knives, providing visual confirmation of detection

4.3.1Figure of Training Results Table

The table displays YOLO's key training metrics across multiple epochs, including box loss, class loss, DFL loss, precision, recall, and mAP. As the epochs progress, all loss values gradually decrease, showing that the model is learning to make more accurate predictions. At the same time, precision, recall, and mAP increase steadily, indicating improved detection performance. Overall, the results reflect a consistently improving and well-converging model.

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4.3.2Figure of Project Folder and Results View

The image shows the full project layout on the left, including dataset folders, model versions, and training outputs. On the right side, the results table is open, allowing you to review how the model performed during training. This setup provides a clear view of both the project organization and the training progress at the same time.

4.3.3Figure of Model Directory (knife_detector_v1)

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The image highlights the knife_detector_v1 directory, showing key files like the model weights (best.pt and last.pt), labeled training images, and the results.csv containing training metrics. These files together represent the trained model and its performance data, ready for testing or further development.

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CHAPTER-5 (OBJECTIVE 2)

5.1 INTRODUCTION

Weapon detection

In today's world, ensuring public safety has become a major concern due to the increasing incidents of violence involving weapons in places such as schools, airports, offices, and public areas. Traditional surveillance systems like CCTV cameras are widely used, but they mainly serve as passive monitoring tools that rely heavily on human operators. Continuous monitoring by humans is prone to fatigue, errors, and delayed responses, which can lead to serious consequences during critical situations.

To overcome these limitations, the proposed Weapon Detection and Alert System using Deep Learning aims to provide an automated and intelligent solution for real-time threat detection. This system utilizes advanced deep learning techniques, particularly the YOLOv8 object detection model, to identify weapons such as guns and knives from live video streams. By analyzing each frame of the video in real-time, the system can accurately detect potential threats and reduce false alarms.

Once a weapon is detected, the system immediately triggers alerts through email and SMS notifications to concerned authorities, enabling quick response and preventive action. The integration of real-time monitoring with an automated alert mechanism enhances the efficiency of surveillance systems and minimizes human dependency. This project plays a crucial role in improving security and safety by providing a fast, reliable, and scalable solution for modern surveillance needs.

Furthermore, the system is designed to work efficiently in real-world conditions such as varying lighting, motion, and crowded environments. It also incorporates optimized processing techniques to ensure high detection speed and low latency. By combining artificial intelligence with surveillance technology, this system contributes towards building smarter and safer environments. The proposed solution can be further extended and integrated with smart city infrastructure for large-scale security applications.

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5.2 PROPOSED METHODOLOGY

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The methodology is divided into several key phases:

5.2.1 Data collection:

Collect images of weapons (guns, knives) and non-weapon objects from sources like Kaggle and Roboflow.

5.2.2 Data Annotation:

Label the images using tools like Labellmg by drawing bounding boxes around weapons.

5.2.3 Data Preprocessing:

Apply techniques like resizing, normalization, and augmentation (rotation, flipping) to improve model.

5.2.4 model Selection:

Use YOLOv8, a deep learning-based object detection model, for accurate and real-time detection.

5.2.5 Model Training:

Train the YOLOv8 model using the prepared dataset to learn weapon features.

5.2.6 Model Testing and Validation:

Evaluate the model to ensure high accuracy and low false positive rate.

5.2.7 Real-Time Detection:

Integrate the trained model with webcam/CCTV using OpenCV to detect weapons in live Images.

5.2.8 Detection Filtering:

Set a confidence threshold to avoid false detections and improve reliability.

5.2.9 Alert Generation:

Trigger alerts when a weapon is detected by sending notifications via Email (SMTP) and SMS

5.2.10 Multi-threading:

Use parallel processing to send alerts without interrupting real-time detection.

5.2.11 Figure (Flow Chart)

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This flowchart outlines the systematic workflow of a real-time computer vision application designed for knife detection. The process begins with the System Initialization, where the environment is prepared using Python, OpenCV, and specific model tools. Once the environment is ready, the AI Model is loaded using pre-trained YOLOv8 weights (specifically a file named best.pt) tailored for identifying weapons. With the software foundation set, the system starts the video feed via a webcam, which then undergoes frame formatting to resize images to 640 × 640 pixels and normalize pixel values for optimal processing.

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5.2.1 This flowchart illustrates a real-time security pipeline that utilizes a pre-trained YOLOv8 AI model to monitor a webcam feed, analyze frames for weapons, and trigger visual alerts upon detecting a knife.

The core of the operation occurs during the AI Analysis phase, where the YOLOv8 model scans each frame to

generate bounding boxes and confidence scores. This leads to a Critical Check that asks: "Is a knife detected?" If the answer is Yes, the system triggers a Detected Alert, overlaying visual indicators and warning text onto the screen. If the answer is No, the system continues to display a normal feed, showing the clean webcam video without any interruptions. This loop ensures continuous monitoring and immediate threat identification.

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5.3 RESULT AND IMAGES

This image demonstrates the real-time application of the knife detection system, featuring a person holding a knife that has been successfully identified by the AI model. A blue bounding box surrounds the hand and the weapon, labeled with the tag "weapon" and a confidence score of 0.55, indicating the system's live analysis and detection capabilities in a real-world setting.

5.3.1 The image shows the AI system in action, successfully identifying a handheld knife with a blue bounding box and a "weapon" label at a 0.55 confidence level.

This image provides a real-world demonstration of the AI detection system, showing a person holding a knife that has been successfully flagged by the software. A blue bounding box accurately encapsulates the hand and the object, while a digital overlay identifies it as a "weapon" with a confidence score of 0.55. The visual serves as a practical example of the "Display DETECTED Alert" stage from the previously described flowchart, confirming the system's ability to recognize potential threats in a live environment.

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This image provides a high-confidence example of the AI system's detection capabilities, showing a close-up of a knife being held in front of a camera. The software has successfully isolated the object within a vibrant green bounding box, identifying it specifically as a "knife" with a perfect confidence score. This level of precision highlights the model's effectiveness in distinguishing the sharp edges and metallic texture of the blade, demonstrating the final "Detected Alert" phase of the security workflow where the system provides a clear, unmistakable visual confirmation of a potential threat.

5.3.2The image displays a high-precision AI detection where a handheld blade is isolated by a green bounding box and labeled as a "knife" with a confidence score.

The object is highlighted by a prominent green bounding box labeled "knife," illustrating the system's ability to provide clear and definitive visual alerts. This snapshot serves as a prime example of the software's accuracy in identifying specific threats within a live security feed

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CHAPTER-6 (OBJECTIVE 3)

6.1 INTRODUCTION

Real-Time monitoring and Alert system

The Real-Time Monitoring and Alert System is designed to continuously observe an environment and detect potential threats instantly. In traditional surveillance systems, cameras only record video, and human operators are required to monitor the footage, which can lead to delays and errors. This limitation makes it difficult to respond quickly to critical situations.

To address this issue, the proposed system integrates deep learning with real-time video processing to automatically detect weapons such as guns and knives. Using models like YOLOv8, the system analyzes live video streams frame by frame and identifies suspicious objects with high accuracy.

When a weapon is detected, the system immediately triggers an alert mechanism by sending notifications through email and SMS to the concerned authorities. This ensures a rapid response and helps in preventing dangerous incidents.

Overall, the real-time monitoring and alert system improves surveillance efficiency, reduces human effort, and enhances security by providing instant and reliable threat detection in various environments such as public places, offices, and institutions.

The system is capable of operating continuously without interruption and can handle multiple video inputs simultaneously. It is designed to perform effectively under different environmental conditions such as low lighting, motion, and crowded scenes. The use of automated detection reduces the dependency on manual monitoring and minimizes the chances of human error.

Furthermore, the integration of real-time processing with an intelligent alert mechanism enhances situational awareness and improves decision-making during emergencies. This system can be deployed in various locations such as schools, airports, banks, and public spaces to strengthen security infrastructure. Overall, it provides a reliable, efficient, and scalable solution for modern surveillance.

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6.2 Proposed Methodology

6.2.1 Video or image Input Capture:

Capture live video or image stream using a webcam or CCTV camera for continuous monitoring.

6.2.2 Frame Extraction:

Convert the video stream into individual frames for real-time processing.

6.2.3 Preprocessing:

Resize and normalize each frame to match the input size required by the detection model.

6.2.4 Model Integration:

Load the trained YOLOv8 model for detecting weapons in the video frames.

6.2.5 Real-Time Detection:

Analyze each frame using the model to identify weapons like guns and knives.

6.2.6 Bounding Box Generation:

Draw bounding boxes around detected objects with labels and confidence scores.

6.2.7 Confidence Filtering:

Apply a threshold value to remove low-confidence detections and reduce false alarms.

6.2.8 Threat Verification:

Check if the detected object qualifies as a weapon based on predefined conditions.

6.2.9 Alert Triggering:

Generate an alert when a weapon is detected in the frame.

6.2.10 Alert Notification:

Send notifications via Email (SMTP) and SMS (Twilio API) to authorities.

6.2.11 Multi-threaded Processing:

Use multi-threading to send alerts without stopping the real-time detection process.

6.2.12 Continuous Monitoring:

Ensure the system runs continuously to monitor the environment without interruption.

6.2.13 Figure (Flow Chart)

This flowchart illustrates the multi-channel escalation process of an Automated Threat Response & Alert System designed for high-security environments. Once a threat is detected and the alert is triggered, the system simultaneously activates three main response branches: an Automated Response Unit, an Instantaneous Local Alert featuring audible alarms, and Remote Authority Notifications via email or police contact

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6.2.1 This diagram outlines an automated security protocol that, once a threat is triggered, simultaneously activates local alarms, notifies authorities, and logs captured evidence for immediate human intervention.

This flowchart illustrates the multi-channel escalation process of an Automated Threat Response & Alert System designed for high-security environments. Once a threat is detected and the alert is triggered, the system simultaneously activates three main response branches: an Automated Response Unit, an Instantaneous Local Alert featuring audible alarms, and Remote Authority Notifications via email or police contact. These automated actions funnel into critical human-centric outcomes, including direct Email/SMS alerts to law enforcement and the mobilization of Human Intervention. To ensure accountability and future investigation, the entire event is logged within Secure Evidence Storage, which captures relevant images and timestamps as the incident unfolds.

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6.3 RESULT AND IMAGES

This image displays an email inbox showing the practical execution of the "Remote Authority Notification" phase of the security system. It features a series of automated "CRITICAL ALERT" emails sent from a dedicated security account, notifying the user that potential threats like knives, weapons, or guns have been detected.

6.3.1 The image showcases automated email notifications sent by the security system to provide real-time critical alerts and photographic evidence whenever a potential weapon is detected.

This image displays an email inbox showing the practical execution of the "Remote Authority Notification" phase of the security system. It features a series of automated "CRITICAL ALERT" emails sent from a dedicated security account, notifying the user that potential threats like knives, weapons, or guns have been detected. Each notification includes a clear subject line (ALERT: Weapon Detected!) and mentions an attached image for immediate visual verification, demonstrating how the system provides real-time, actionable data to authorities or security personnel.

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CHAPTER 7

Tools And Technologies

7.1 Platform

The system is developed using a combination of software-based platforms and edge computing techniques. Python is used as the primary programming language along with the YOLOv8 framework for implementing real-time weapon detection. The system runs on platforms like Windows or Linux and can be deployed on edge devices such as Raspberry Pi for lightweight and real-time performance. These technologies enable efficient video processing, AI inference, and real-time alert generation within a unified system.

7.2 Development Tools

7.2.1 Python

Python is the primary language used for developing the weapon detection system. It provides extensive libraries for machine learning, computer vision, and real-time processing. Python is used for training the YOLOv8 model, preprocessing datasets, and performing real-time detection using camera input.

7.2.2 OpenCV

OpenCV is used for capturing and processing real-time video streams from webcams or CCTV cameras. It helps in frame extraction, image processing, and displaying detection results with bounding boxes.

7.2.3 YOLOv8

YOLOv8 is a deep learning-based object detection model used for identifying weapons in images and videos. It provides high accuracy and fast detection, making it suitable for real-time surveillance applications.

7.2.4 NumPy and Pandas

NumPy and Pandas are used for data manipulation and handling datasets. They assist in preprocessing, organizing, and managing training data efficiently.

7.2.5 Annotation Tools (LabelImg / Roboflow)

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These tools are used to label images by drawing bounding boxes around weapons. Proper annotation helps improve model training and detection accuracy.

7.2.6 SMTP (Email Service)

SMTP protocol is used to send email alerts when a weapon is detected. It ensures that authorities receive instant notifications with captured images.

7.2.8 Development Environment

Tools like VS Code or PyCharm are used for writing and debugging the code. They provide an efficient interface for project development.

7.2.9 Version Control (Git)

Git is used to manage code versions and track changes during development. It helps in collaboration and maintaining project history.

7.3 Data Processing and Analysis

7.3.1 Data Collection:

Collect images of weapons (guns, knives) and non-weapons from sources like Kaggle and Roboflow.

7.3.2 Data Cleaning:

Remove duplicate, blurred, and irrelevant images to improve dataset quality.

7.3.3 Data Annotation:

Label images using tools like LabelImg by drawing bounding boxes around weapons.

7.3.4 Data Preprocessing:

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Resize images, normalize pixel values, and prepare data for model input.

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7.3.5 Data Augmentation:

Apply techniques like rotation, flipping, and scaling to increase dataset diversity.

7.3.6 Dataset Splitting:

Divide data into training, validation, and testing sets

7.3.7 Model Training:

Train the YOLOv8 model using the processed dataset to learn weapon features.

7.3.8 Model Evaluation:

Analyze performance using metrics like accuracy, precision, recall, and F1-score.

7.3.9 Testing:

Test the model on unseen data to verify its real-world performance.

7.3.10 Real-Time Frame Analysis:

Process each video frame to detect weapons using the trained model.

7.3.11 Confidence Thresholding:

Filter detections based on confidence score to reduce false positives.

7.3.12 Result Analysis:

Monitor detection results and system performance for improvements.

7.4 Embedded Communication and Control

7.4.1 Microcontroller Integration:

A microcontroller (such as Arduino or Raspberry Pi) is used to control system operations and manage hardware components.

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7.4.2 Sensor and Camera Interface:

The system connects with cameras (webcam/CCTV) to capture real-time video input for processing.

7.4.3 Data Communication:

Communication between components is handled using serial communication protocols or network-based communication.

7.4.4 Real-Time Processing Control:

The embedded system ensures continuous data flow and real-time execution of detection tasks.

7.4.5 Alert Communication Module:

Integrates communication modules (Wi-Fi/GSM) to send alerts via Email (SMTP) and SMS (Twilio API).

7.4.6 Control Logic Implementation:

Decision-making logic is implemented to trigger alerts only when a weapon is detected with high confidence.

7.4.7 Multi-threading Support:

Enables parallel execution of detection and alert systems without interruption.

7.4.8 System Reliability:

Maintains stable communication and control for uninterrupted monitoring and alerting.

7.5 Machine Learning and Real-Time Inference

7.5.1 Model Selection:

Use YOLOv8, a deep learning-based object detection model, for identifying weapons.

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7.5.2 Dataset Training:

Train the model using annotated images of weapons and non-weapon objects.

7.5.3 Feature Learning:

The model learns patterns, shapes, and features of weapons during training.

7.5.4 Model Optimization:

Fine-tune parameters to improve accuracy and reduce false positives.

7.5.5 Model Deployment:

Load the trained model into the system for real-time usage.

7.5.6 Live Video or Image Input:

Capture real-time video from webcam or CCTV camera.

7.5.7 Frame-by-Frame Processing:

Convert video into frames and analyze each frame individually.

7.5.8 Real-Time Inference:

Apply the trained model on each frame to detect weapons instantly

7.5.9 Bounding Box Detection:

Draw boxes around detected weapons with labels and confidence scores.

7.5.10 Continuous Monitoring:

Keep the system running continuously for real-time surveillance and quick response.

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CHAPTER 8

Software Testing And Results

8.1 Unit Testing

Individual components like model loading, video capture, and alert functions are tested separately. Ensures each module (YOLOv8 detection, email sending, SMS sending) works correctly.

Scope:

Testing individual components like YOLOv8 model loading, video capture, and alert functions separately. Ensures each module works correctly without errors.

Tools Used:

Python (unittest module)

VS Code / PyCharm (for debugging)

OpenCV (for testing video input)

8.2 Integration Testing:

Combined modules such as detection + alert system are tested together. Verifies that the model correctly triggers email/SMS alerts after detecting a weapon.

Scope:

Testing interaction between modules such as detection + alert system.

Ensures smooth data flow from weapon detection to alert generation.

Tools Used:

Python scripts

OpenCV (for real-time integration testing)

SMTP (for alert verification)

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8.3 System Testing:

The complete system is tested as a whole using real-time video input. Ensures end-to-end functionality including detection, processing, and alert generation works properly.

Scope:

Testing the complete system with real-time video input.

Verifies overall performance, accuracy, and alert response.

Tools Used:

Webcam / CCTV camera

YOLOv8 model

OpenCV

Email (SMTP) & SMS

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CHAPTER 9

CO-PO-PSO Mapping

CO-PO-PSO ²¹ PO1 PO2 PO3 PO4 PO5 PO6 PO7 PO8 PO9 PO10 PO11 PSO1 PSO2

CO1: Identify
and formulate the
problem
statement by
applying
engineering
knowledge and
problem analysis
techniques.

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CO2: Summarize
an exhaustive
literature review
in the field of the
proposed work
and identify
research gaps.

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CO3:
Demonstrate
sound technical
knowledge of the
selected project
topic through
implementation
and analysis.

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The mapping of CO-PO-PSO to the project “ Weapon Detection And Alert System using Deep Learning”

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CHAPTER 10

Conclusion And Future Work

10.1 Conclusion

The weapon detection and alert system using deep learning successfully demonstrates the use of advanced artificial intelligence techniques to enhance public safety and surveillance systems. By utilizing the YOLOv8 model, the system is capable of accurately detecting weapons such as guns and knives in real-time video streams with high efficiency and low latency.

The integration of real-time monitoring with an automated alert mechanism ensures immediate notification through email and SMS, enabling quick response to potential threats. The system also minimizes false positives by using a well-prepared dataset that includes both weapon and non-weapon objects.

The system performs effectively under different environmental conditions such as varying lighting, motion, and crowded scenes. It is designed to provide continuous monitoring without interruption, ensuring reliable performance in real-world scenarios. The use of multi-threading further improves system efficiency by allowing detection and alert generation to occur simultaneously.

In addition, the system is flexible and can be easily adapted to detect other types of objects by retraining the model with different datasets. It can also be integrated with existing surveillance infrastructure, making it cost-effective and practical for deployment in various environments.

Overall, the proposed system reduces dependency on manual surveillance, improves detection accuracy, and provides a reliable and scalable solution for modern security applications. This project highlights the potential of deep learning in building intelligent surveillance systems that can contribute to safer environments in public and private spaces.

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10.2 Future Work

10.2.1 Dataset Expansion:

Include more weapon types and diverse real-world images to improve accuracy.

10.2.2 Low-Light Enhancement:

Improve detection in poor lighting and night-time conditions.

10.2.3 Model Optimization:

Use technologies like TensorRT or OpenVINO for faster inference.

10.2.4 IoT Integration:

Connect the system with smart devices for large-scale monitoring.

10.2.5 Audio Detection:

Integrate sound detection (e.g., gunshot detection) for better accuracy.

10.2.6 Multi-Camera Support:

Extend the system to monitor multiple cameras simultaneously.

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